



Research article

Key information technology and management issues 2010–11: an international study

Jerry Luftman¹, Hossein S Zadeh²

¹Howe School of Technology Management, Stevens Institute of Technology, Hoboken, USA;

²School of Business IT & Logistics, RMIT University, Melbourne, Australia

Correspondence:

J Luftman, Howe School of Technology Management, Stevens Institute of Technology, 1 Castle Point on Hudson, Hoboken, NJ 07030, USA.

E-mail: Jerry.Luftman@stevens.edu

Abstract

The importance of the impact of IT for organizations around the world, especially in light of the global financial crisis, has amplified the need to provide a better understanding of the specific geographic similarities and differences of IT managerial and technical trends. Going beyond identifying these influential factors is also the need to understand the considerations for addressing them, in light of recognizing the respective local characteristics, especially when operating in a globally linked environment. By comparing and contrasting different geographies, this paper presents important local and international factors (e.g., management concerns, influential technologies, budgets/spending, organizational considerations) necessary to prepare IT leaders for the challenges that await them. The research is based on data from four geographic regions (United States, Europe, Asia, and Latin America). The same questionnaire (although translated for the respective respondents), based on the lead authors of the well-respected and long-running Society for Information Management survey, was applied across geographies. This paper presents the major findings based on survey responses from 472 organizations (172 US, 142 European, 103 Asian, and 55 Latin) in mid-2010. The top five management concerns were: (1) business productivity and cost reduction; (2) IT and business alignment; (3) business agility and speed to market; (4) business process re-engineering; and (5) IT reliability and efficiency. The five most influential technologies were business intelligence, cloud computing, enterprise resource planning, Software as a Service/Platform as a Service, and collaborative tools.

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Introduction

The Society for Information Management (SIM) has been conducting a survey of senior IT executives since 1980 to help IT and business leaders understand important issues and trends. There are a number of IT trends survey results reported in the literature, but most are either specific to a certain industry, such as IT (Herlihy and Ptak, 1997) and airlines (Airline Business, 2009), or concentrate on a specific aspect of an industry, such as IT salaries (Ziff Davis Enterprise, 2011). There have been a limited number of surveys covering various aspects of the IT industry (Herlihy and Ptak, 1997; O'Toole, 2003; Baker,

2006; Airline Business, 2009; Ziff Davis Enterprise, 2011), most with limited geographic reach (Li *et al.*, 2005). The SIM survey remains one of the longest-running surveys covering various aspects of IT industry. Organizations around the world have used the findings of the previous SIM surveys as a barometer to prepare for the future, although the annual SIM survey has in the past been based on survey responses just from organizations located in the United States.

The recognition of the global reach of IT, especially in light of the impact of the global financial crisis, has amplified the necessity to obtain responses from organizations

Table 1 Percentage of Respondents by Geography

<i>Geography</i>	<i>Total</i>
United States	172
Western Europe	142
South East Asia	75
Australia	28
Latin America	55
TOTAL	472

Table 2 Percentage of Respondents by Industry

<i>Industry classification</i>	<i>Percentage</i>
IT Services/vendors	13.4
Finance/Banking/Insurance	13.4
Manufacturing/Process	13.4
Computer/Network Consulting	8.5
Business Services/Executive Search	8.0
Education/Publishing	7.8
Wholesale/Retail/Distribution	6.3
Construction/Utilities/Engineering	4.9
Pharmaceutical/Healthcare	4.6
Network Communications	4.4
Transportation	4.0
Government	3.4
Real Estate/Legal	3.0
Travel, Hospitality/Entertainment/Sports	2.9
Non-Profit	2.0

around the globe to understand similarities and difference across geographies. Hence, this year the same survey was conducted in three major geographies outside of the US, namely, Europe, Asia, and Latin America.

This paper presents the major insights gained from this year's survey in Europe, South East Asia (SE Asia), Latin America, and the United States. This paper is based on responses from IT executives representing over 472 organizations (see Tables 1 and 2 for participant demographics).

While the findings from the US (with some minor comparison to Europe and SE Asia/Australia) have been published in the December 2010 issue of *MISQ-Executive* (*MISQE*), this paper, using an updated set of data, includes the major insights gained from Europe, SE Asia, and Latin America, and uses the overall global responses as the benchmark for comparison purposes.

Research methodology

In the summer of 2010, a survey of senior IT executives from over 472 organizations located in the United States, Europe, Asia, and Latin America was conducted. Table 1 provides a breakdown of respondents based on their geography and Table 2 provides a breakdown of the respondents by industry. While the December 2010 issue of *MISQE* presented the US data in detail, this paper focuses on analyzing the geographies outside of the US. Additional methodological information is provided in the appendix.

The four categories focused on in the 2010 survey were:

- (a) Top IT management concerns
- (b) Top application and technology investments
- (c) IT budget allocation
- (d) IT organizational considerations

Participants were asked to rate the importance of 31 managerial concerns, 52 application and technology investments, 10 budgetary considerations, 3 staffing and remuneration matters, and 9 organizational considerations.

The top 10 management concerns across the geographies were aggregated to provide the global top 10 management concerns. The global top 10 management concerns along with the respective ranking for each of the items in the various geographies are presented in Table 3. These concerns are elaborated on throughout this paper.

Similarly, the top 10 application and technology development tools were aggregated to provide a global top 10 application and technology development tool list. The global list and the corresponding ranking for each of the tools in various geographies are presented in Table 4 and are elaborated on in this paper.

A. Top 10 IT management concerns

The top 10 US management concerns tend to evolve slowly, although the severe recession begun in 2008 impacted some of the management priorities in 2009. Four US concerns that have traditionally remained on the top 10 list are: IT and business alignment, business process management and reengineering, IT strategic planning, and security and privacy (Luftman and Ben-Zvi, 2010b). Three of these four persistent concerns appear in the top 10 lists from all of the geographies; security and privacy was ranked 20th by Europe, 11th by SE Asia, and 9th by both the US and Latin America. Other differences in the top 10 include IT cost reduction being ranked outside of the top 10 in both SE Asia and Latin America; project management being ranked outside of the top 10 by only the US and Europe; and globalization being ranked 15th and 13th by European and Latin American IT managers, respectively.

Interestingly, during this economic downturn, IT cost reduction ranked 14th in SE Asia, where the prognosis for the financial sector is relatively better compared to other emerging economies (*MoneyWeek*, 2009; Plessis, 2009; Ryan, 2009), as opposed to the US (ranked 8th), Europe (ranked 5th), and Latin America (ranked 11th), where the economic conundrum is on the front pages of daily newspapers. The discussions below elaborate on this finding further. The numbers represent the overall importance of the ranking for the concern.

1. Business productivity and cost reduction

Business productivity and cost reduction appears as the top concern in all geographies (ranked first universally). It has only recently been gaining momentum (it first appeared in the SIM survey in 2007), but it now seems to dominate the top management concerns globally.

The IT cost reduction (discussed above), and the importance of improving business productivity is in line with the findings of Luftman and Ben-Zvi (2010b) that the current trend seems to be unique in this recession; instead of

Table 3 Top IT Management Concerns

	<i>Global</i>	<i>US</i>	<i>Europe</i>	<i>SE Asia (inc Australia)</i>	<i>Latin America</i>
Business productivity and cost reduction	1	1	1	1	1
IT and business alignment	2	3	3	2	3
Business agility and speed to market	3	2	4	6	2
Business process management and reengineering	4	5	2	3	4
IT reliability and efficiency	5	4	6	4	8
Revenue-generating IT innovations	6	7	7	6	7
IT strategic planning	7	6	7	9	5
IT cost reduction	8	8	5	14	11
Project management	9	16	18	6	6
Globalization	10	10	15	10	13
Security and privacy	11	9	20	11	9
Enterprise architecture	12	13	11	12	14
Considerations of IT human resources	13	14	16	5	16
Change management	14	11	11	16	15
Knowledge management	15	15	17	12	10
Supplier management/Outsourcing	16	12	7	18	19
Leadership role of the CIO	17	20	11	19	12
Sourcing decisions	18	17	7	20	20
Social implications of IT	19	19	19	17	17
Design in the IT area	20	21	21	14	18

Table 4 Top Application and Technology Development

	<i>Global</i>	<i>US</i>	<i>Europe</i>	<i>SE Asia (inc Australia)</i>	<i>Latin America</i>
Business intelligence	1	1	3	2	2
Cloud computing	2	5	1	3	3
Enterprise resource planning (ERP) systems	3	3	—	1	4
Software as a Service (SaaS, PaaS)	4	—	2	—	5
Collaborative and workflow tools	5	—	—	—	1
Business process management systems	6	—	4	5	—
Virtualization	7	2	—	—	—
Continuity planning/disaster recovery	8	4	—	—	—
Microsoft upgrade	9	—	—	4	—
Security	10	—	5	—	—

simply cutting IT budgets, IT leaders seem to be responding to this recession by focusing on IT as an enabler/driver of business productivity for the rest of the business. Even though data for individual countries are not reported in this paper, it is interesting to note that this trend is present in all geographies except in Australia, where issues such as IT strategic planning (and hence the role of CIOs) are ranked higher than business productivity and cost reduction. Since Australia is the only western country that did not undergo a recession in 2008/2009 (*MoneyWeek*, 2009; Plessis, 2009), placing more focus on strategic issues instead of cost reduction is understandable. It is still noted that business productivity and cost reduction rank very high in Australia, but they are simply not their top management concern.

2. IT and business alignment

Alignment of IT and business continues to be elusive, and in all of the geographies it ranks in the top 10 management

concerns, ranking 3rd in the US, Europe, and Latin America, and even higher at the 2nd place in SE Asia.

IT leaders in most demographics see IT as an integral driver/enabler of efficiency/effectiveness in other parts of the business and therefore focus on initiatives that enhance the maturity of alignment between IT and business. While IT business alignment has been recognized for over 30 years as a persistent problem, it is clear that it is universal.

3. Business agility and speed to market

Business agility and speed to market is ranked No. 2 in the US and Latin America, 4th in Europe and 6th in SE Asia. Business agility and speed to market are essential for business survival in an uncertain and volatile economy, and the improved ranking of business agility and speed to market is testimony to that. The US results demonstrate this well because in the past couple of years business agility and speed to market has moved up from the mid-teens to

the No. 2 management concern. However, as discussed earlier, the impact of the recession has been subdued in SE Asia, and as expected business agility and speed to market ranks relatively lower in that area (6th). It is noted that business agility and speed to market, along with business productivity and cost reduction, are the foundation for long-term competitive advantage, and therefore it is anticipated that these concerns will improve their ranking in SE Asia as these geographies will try to maintain and improve their global competitive advantage.

4. Business process management and reengineering

Business process management (BPM) and business process reengineering (BPR) are ranked in the list of top 10 concerns in all of the geographies (ranked 2nd, 3rd, 4th, and 5th in Europe, SE Asia, Latin America, and the US, respectively). Since the 1990s large corporations leveraged the BPR tools that analyze design, and automate workflows and processes within an organization, and as a result BPM and BPR have appeared every year in the list of top 20 concerns in the US.

BPR by definition is 'process-centric'. More recently, BPM has emerged as a more holistic approach focusing on integrating all aspects of the organization. It has become an important tool to take advantage of BPR initiatives. BPM is utilized to streamline end-to-end management of the whole enterprise (enterprise-centric). During the economic downturn, as organizations focus on leveraging IT to reduce expenses by enhancing business processes and as the recovery from the recession gets underway, corporations large and small need to compete in a globally linked marketplace, and it is expected that business process management and reengineering will remain a top management concern globally. The high ranking of enterprise resource planning (ERP) as an important application and technology (discussed later in this paper) provides further support for this important consideration.

5. IT reliability and efficiency

The growing complexities of IT systems, along with the need to consider costs, have brought IT reliability, availability, and efficiency to the forefront. IT reliability and efficiency is ranked 4th in the US and SE Asia, 6th in Europe, and 8th in Latin America. This management concern has been rising in the US since 2008, when it rose from 8th place in 2008 to 6th in 2009 and 4th in 2010 (Luftman and Ben-Zvi, 2010b). The only geography that ranks IT reliability and efficiency as high as the US is SE Asia.

IT reliability and efficiency refers to the accuracy, timeliness, and reliability of data and information delivered by IT (Luftman and Ben-Zvi, 2010b). The relatively higher ranking of IT reliability and efficiency in Latin America could have two distinct explanations. It could be that to have a reliable and efficient IT, stored data and information need to be pristine, and that Latin American IT leaders might be concerned about the quality and accuracy of data/information in their organizations. On the other hand, the ranking could be because Latin American organizations tend to favor IT strategic planning and project management more than other geographies.

After Amazon pulled the plug on Wikileaks' Web hosting service (McLeod, 2010), leaving the whistle-blowing website unavailable for a couple of days, which was followed by Domain Name Server issues for the website in the US, Wikileaks had to move its Web hosting servers to Sweden, then France, and finally to Switzerland (as reported by Bloomberg (Campbell, 2010) and Netcraft (News.Netcraft.com, 2010)). Without prejudice regarding political, ethical, and social aspects of the Wikileaks' recent release of classified information, the developments surrounding its Web hosting issues have been subject of much discussion regarding IT reliability and efficiency, as well as security (discussed below). With Wikileaks' website currently hosted in Europe, we anticipate IT reliability and efficiency, in addition to security, becoming much more important management concerns in the next 1 or 2 years.

6. Revenue-generating IT innovations

Revenue-generating IT innovations tied for 7th in the US, Europe, and Latin America, and 6th in SE Asia, but only 17th in Australia. In the US, revenue-generating IT innovations has moved up to tie for No. 6 in 2010, from 8th in 2009, and from 17th in 2008. This concern was first included in the SIM survey in 2007. These rankings, without exception, clearly match the economic outlook in the respective geographies because during the recession, executives searched for alternative ways of generating revenues, and one way is through IT innovations. In all geographies, revenue-generating IT innovations has ranked in the top 10, except in Australia, where the economic outlook has been better than that of the other areas.

7. IT strategic planning

IT strategic planning was the top ranking concern in the US in 1980, and has remained in the top 10 ever since, ranking 6th in the US in 2010; European, SE Asian, and Latin American IT leaders ranked it 7th, 9th, and 5th, respectively. An important consideration, although, perhaps, not as critical during an economic downturn.

8. IT cost reduction

IT cost reduction remains a concern for all IT managers, although with a direct correlation to the local economic outlook. The importance of IT cost reduction is more or less similar in the US (at 8th) and Europe (5th), but it falls outside of the top 10 management concerns in SE Asia and Latin America (at 14th and 11th places, respectively). With the global economy slowly recovering from the recession, budget restrictions might be easing, but are still not what they were prior to the recession. Infrastructure improvements and utilization of technologies such as cloud computing (discussed in Top Technologies section) are being used by European, SE Asian, and Latin American IT leaders to curb spending. Virtualization is ranked very low in all demographics except in the US.

In the global IT trade, exchange rates play an important role, as both the actual trade and the exchange of the money are done in real time. A cursory browse of the exchange rate of various currencies against the US dollar (USD) clearly demonstrates why IT cost cutting ought to be of much higher concern in SE Asia, than it is in Europe and Latin

America. SE Asian currencies (with the exception of those currencies pegged to the USD, such as the Chinese yuan and Hong Kong dollar), as well as the Australian dollar, are trading at or close to record highs. For example, the Singaporean dollar has increased against the USD over the past 2 years from a low of 0.64 to a high of 0.77, and is trading at 0.76 as of this writing (XE.com, 2010d), an increase of over 20% in 2 years. Similarly, the Indian rupee has increased from a low of 0.019 of 2 years ago to the current high of 0.022 (15.7% increase) (XE.com, 2010b), and the Australian dollar has increased from a low of 0.62 against the USD around 2 years ago to over parity now (an increase of over 60%). These substantial increases mean all commodities and services (including IT products and services) originating from SE Asia are globally much more expensive now than they were 2 years ago.

On the other hand, the euro and Latin American currencies do not show the same appreciation against the USD, translating to European and Latin American products and services retaining their price competitiveness in the global market. The euro has had a low of 1.19 and a high of 1.5 over the past 2 years, and is currently trading at 1.3 against the USD (XE.com, 2010a), and Mexican pesos showing a nominal increase from 0.075 to around 0.08 over the past 18 months (Xe.com, 2010c).

The pressure to keep the USD low (Lien, 2010) directly translates to high SE Asian currencies, but Europe is having its own reasons to keep the euro low (WallStreetJournalOnline, 2010). In the unlikely event that the euro starts climbing against the USD, IT cost cutting will become a major concern in the coming year in Europe.

Our prognosis is that in areas where the local currency remains at or near record highs against the USD (such as SE Asia and Australia), there remains an increasing pressure on IT managers to cut IT costs to retain their global competitiveness, which will mean increased ranking of IT cost cutting as a management concern. However, in areas where the exchange rates do not show any out-of-ordinary movements against the USD (such as in Europe and Latin America), IT cost cutting will still remain a management concern, albeit not increasing its ranking markedly.

9. Project management

Project management was ranked among the top 10 management concerns in all of the areas, except in the US and Europe, where well-established markets and a much longer history of successful IT project completions have pushed project management much further down the list of concerns. Increased experience in successful completion of projects on the one hand, and with a constant stream of bad publicity surrounding failed IT projects on the other, it is predicted that level of concern toward project management to merge to similar level in different geographies. It currently stands as the 6th management concern in SE Asia and Latin America, and 16th and 18th in the US and Europe, respectively.

10. Globalization

The evolution of supply, demand, global market trends, and environmental factors is driving companies toward operating as if they reside in a homogeneous worldwide market

(Gamboa, 1988). In the 1980s and 1990s, governments from all of the major developed countries were advocating free trade and lower trade barriers. However, since the start of the recent recession, a more protectionist approach has been adopted to curb rising unemployment and balance trade deficits. In fact, all G-20 countries introduced protectionist measures in some shape or form (Hufbauer *et al.*, 2010). However, many of these measures have had an adverse effect on jobs (Hufbauer *et al.*, 2010). As recently as September 2010, Patricia Francis, Executive Director of the UN International Trade Center, called on global economic policy makers to oppose and curb trade protectionism (*GlobalTimesChina*, 2010). In this context, it is no wonder that globalization ranks in the list of top 10 management concerns in most areas (7th in Australia and Latin America and 10th in the US, but 12th in Europe, 15th in SE Asia).

As IT is believed to be an essential tool to revitalize cross-border trade and revive the world economy in the wake of the recession (Chiew, 2009), our prognosis is that these protectionist measures will be reduced, or scrapped altogether, in the next few years, prompting a big jump in global trade in general, and in global IT trade in particular.

B. Top five applications and technology investments

Survey respondents were asked to rank the importance of applications and technology developments by selecting their top five from a list of 52 items. Similar to the SIM survey, respondents were also asked to nominate new technologies if it was not already in the list. Table 4 lists the top 15 application and technology rankings for 2010 across the geographies. The top five applications and technologies vary greatly across the geographies, and this is indicative of the different cultural, technological, infrastructure, and human resource variation in those areas. The benchmark top five applications and technologies for 2010 are discussed below and comparisons are made with the target geographies.

1. Business intelligence

European companies ranked business intelligence as their 3rd top application and technology development, and SE Asian and Latin Americans as their 2nd. It has also been ranked the highest in the US, 1st in 2009 and 2010, and 2nd in 2006, 2007, and 2008 (Luftman and Ben-Zvi, 2010a,b). Since business intelligence leverages data mining to identify valuable insight, this high ranking across these geographies suggests that IT leaders believe their organizations are data rich and insight poor (Luftman and Ben-Zvi, 2010b). Especially in places such as SE Asia where overwhelming rapid growth in the past decade may have prevented pristine data integration and integrity, business intelligence might not be quite as effective as in the other geographies.

2. Cloud computing

Cloud computing ranks 3rd in both SE Asia and Latin America, and 5th in the US, while it tops the chart in Europe. It is noted that cloud computing was a very recent entrant in the SIM survey appearing as No. 17 in 2009. Even though the implications of cloud computing are yet to be fully understood, in the short term it allows organizations to look for appropriate skills and expertise beyond their

immediate vicinity. General shortage of skilled labor in SE Asia (Nystedt, 2010) and the shortage and the expense of skilled labor in some part (Beer, 2010; Herrick, 2010) can be major drivers of cloud computing in that region. Shortage and cost of skilled labor is less of an issue in the US, where there is a reasonable number of skilled labor available at reasonable costs.

The availability of inexpensive and clean power is another driving factor behind adopting cloud computing. In a research at Lawrence Berkeley National Laboratory, Jonathan Koomey reported aggregate electricity use for data centers doubled worldwide from 2000 to 2005 (Koomey, 2008). Furthermore, he reported 'total electricity use grew at an average annual rate of 16.7% per year, with the Asia Pacific region (without Japan) being the only major world region with growth significantly exceeding that average'. The electricity cost of data centers in Australia, for example, are expected to grow by 60% over the next 2 years (Corrigan, 2010).

In the absence of any prospect for a major upgrade of electricity production and distribution grid in many developed countries, cloud computing will remain an increasingly attractive proposition for IT managers. In addition, with the continued low priority placed on IT infrastructure expenditure planned for the next year because of the economy (see next section), cloud computing will remain an important priority, as will its close cousin, virtualization.

3. ERP systems

It is no surprise that the previously discussed high ranking of the management concern BPM/BPR would bring with it commensurate high rankings for the associated applications and technologies such as ERP and workflow tools, in SE Asia (1st for ERP), Latin America (1st for collaborative tools), and the US (3rd for ERP). See the Collaborative and workflow discussion below.

4. Software as a Service

Software as a Service (SaaS) and Platform as a Service (PaaS) are delivery of software solutions or computing platforms as a service across computer networks. They are very related to cloud computing. In a densely populated region with diverse historical, cultural, and linguistic backgrounds such as Europe, SaaS and PaaS make the most sense, as it is close to impossible for small players to cater for all diverse types of clients. For example, it is more difficult to run a commerce server in multiple language and currencies (in Europe) than it is to run a single-language, single-currency commerce site in the US. The high ranking of SaaS and PaaS at No. 2 in Europe is a testimony to this. Similarly, the same technology ranks as the 4th in Latin America. It, however, does not rank in the top 10 in the US and SE Asia. We envisage SaaS and PaaS to remain in the list of top 10 tools in Europe and Latin America, and to improve in their ranking in the other geographies, especially SE Asia due to the reason stated above.

5. Collaborative and workflow tools

Collaborative and workflow tools are ranked as the top application and technology development in Latin America, as IT managers try to embrace highly successful social

media tools into workflows. On the other hand, these tools appear only in the 20th spot in SE Asia as online collaborative tools in the workplace are relative newcomers in that area and language barriers have traditionally limited mass adoption of tools such as Facebook and Twitter. Facebook, for example, is expanding extraordinarily in SE Asia. For example in the US, it has expanded 141% from December 2008 to December 2009, but in Indonesia, Philippines, and Thailand it has expanded by more than 1500%, 2000%, and 1000%, respectively (Burcher, 2009). As social networking matures, it is expected that enterprises will take advantage of the private use of these tools, and therefore it is expected that ERP and collaborative and workflow tools improve in their ranking in SE Asia and remain important in all the geographies.

6. Side note on top applications and development tools

It would be remiss if we do not cover continuity planning and disaster recovery in an uncertain world. Continuity planning and disaster recovery have become an increasingly important management concern (it ranks at 4th in the US). It is generally the CIO's responsibility to identify the organization's exposure to internal and external threats and to provide effective prevention and recovery plans for the organization.

Major disasters of the past few years (such as the 2004 Indian Ocean tsunami and 2010 Haiti earthquake) have demonstrated that many businesses are simply not prepared for disasters. Worse yet is the fact that many businesses are so unprepared that they cannot survive post disaster. In addition to natural disasters, IT infrastructures are open to attacks such as distributed denial of service (DDoS). The recent DDoS attacks against WikiLeaks, and the ensuing retaliatory DDoS attacks that brought down the websites of major corporations such as PayPal and Mastercard (DarkNet.org.uk, 2010; IslandCrisis.net, 2010), are prime examples that IT infrastructure is susceptible to attacks from adversaries far and beyond the close vicinity of the enterprise.

Continuity planning and disaster recovery is ranked as the 4th management concern in the US, but it surprisingly does not rank in the top 20 concerns in any of the other geographies. Owing to the highly publicized DDoS attacks mentioned above, followed by continuity plan services that have just started to appear in the market (Pauli, 2010), it is anticipated that continuity planning and disaster recovery are becoming much more dominant in the list of management concerns in other areas in 2011.

C. IT budget allocation

This section discusses the survey findings related to the overall allocation of IT budgets with a further discussion on staffing and compensation matters (Table 5).

1. Overall budget allocation considerations

It seems that the rebound from the recession is underway, albeit at different rates in different geographies. CIOs reported increased IT budgets in 2010 compared to 2009 in 34% of the companies in Europe and the US, 51% in SE Asia, and 69% in Latin America. 2011 looks even better with

**Table 5** 2010 IT Budget Allocation (Actual)

	<i>Global (average) (%)</i>	<i>US (%)</i>	<i>Europe (%)</i>	<i>SE Asia (inc Australia) (%)</i>	<i>Latin America (%)</i>	<i>Australia (%)</i>
Hardware, network, software, depreciation	37	32	35	40	42	
Internal staff – domestic	33	43	25	32	30	
Consulting services	8	10	12	4	4	21
Outsourced staff – domestic	10	7	20	7	7	
Outsourced staff – offshore	4	5	5	4	1	
Internal staff – offshore	7	3	3	5	16	

29% of European, 38% of the US, 57% of SE Asian, and 67% of Latin American respondents indicating that their IT budgets would increase in 2011. The anticipated budget increase in SE Asia and Latin America is similar to that of pre-recession budgets in the US, which based on previous SIM surveys, was around 75%.

Looking at the data from another perspective, 85% of Europeans, 82% of SE Asians, 89% of Latin American, and 77% of the US respondents do not believe that their 2011 IT budget will be less than that of 2010. This is indeed on par with the pre-recession figures in the US.

2010 infrastructure expenditures, which in the US have been steadily decreasing during the recession, are 35% of the total IT budget in Europe, 40% in SE Asia, 42% in Latin America, and 32% of total IT budgets in the US.

The projected IT budgets for 2011 show a drop in infrastructure expenditure in Europe (34% in 2011 vs 35% in 2010), Latin America (33% vs 42%), and the US (31% vs 32%). The only geography to increase expenditure on IT infrastructure is SE Asia, where it is expected to slightly rise to 41% in 2011 compared to 40% in 2010. It seems that SE Asian companies are investing in their own infrastructure, perhaps to catch up, while the US and Latin American IT leaders (and to a lesser extent, Europeans) are hedging their bets on getting others to run their systems (see discussion on IT outsourcing), as well as focusing on technologies such as virtualization and cloud computing.

When reviewing the budget allocation for running existing systems vs building/buying new systems, the US has consistently two-thirds existing, one third for new. Europe is about the same, while Asia and Latin America are around 55% for running existing systems vs 45% for building/buying new systems.

2. IT Budget as a percent of revenues

While the often-benchmarked average IT budget as a percentage of revenue for the US was 3.87 in 2010 (relatively the same as in the previous SIM surveys), it was 3.59 in Europe, 3.47 in Latin America, and 5.8 in SE Asia. The somewhat larger number for SE Asia is likely due to the desire to catch up with other parts of the world (see discussion below). An examination by industry reveals that some sectors, such as business services, entertainment/sports, banking/finance, and education/publishing, have IT budgets of more than 5% of their revenue. On the other hand, sectors with IT budgets that are less than 2.5% of their revenue are manufacturing, real estate/legal,

construction, retail, and utilities. The industry breakdown is comparatively consistent across geographies.

3. IT staffing and compensation matters

It is clear that staffing (internal and external) remains the largest component of IT budgets (65% in Europe, 48% in SE Asia, 54% in Latin America, and 68% in the US).

The domestic sourcing budgets for internal staff are 25% in Europe, 39% in SE Asia, 30% in Latin America, and 43% in the US. Conversely, many enterprises in Latin America prefer to take their internal staff through offshoring, accounting for 16% of their total IT budget. Offshore internal staff account for only around 3% of the IT budget in the US and Europe, and a mere less than 0.5% of IT budget in SE Asia. A discussion of the budget allocation for IT outsourcing is included in the outsourcing subsection of the IT organization section.

When it comes to IT staff salaries, SE Asia seems to be the most buoyant geography, showing 64% and 77% of workers having salary increase in 2010 and 2011, respectively, compared to the previous year. This is at the pre-recession levels of salary increases in the US (78% in 2008). Forty-five percent of Latin American IT workers had a pay increase in 2010, and again 45% expect pay increases in 2011. In another sign of recovery from the recession, the US market has seen the most dramatic change, registering salary increases for 42% of IT workers in 2010, and 61% in 2011.

European salary increases bucks the trend with less IT workers registering salary increases in 2011 (48%) than in 2010 (41%). Early 2010 the European Court of Justice ruled that a German law limiting the use of Polish workers was discriminatory (Slegers, 2010). This ruling along with an active recruitment of skilled workers from the neighboring countries (TheNews.pl, 2010) has resulted in an influx of skilled migrants, pushing downward pressure on salary increases in Germany. This, however, might be a temporary influence as the new migrants settle in and demand similar salary and benefits as other IT workers in Western Europe.

The respondents were also asked to indicate the anticipated percentage of their IT budgets allocated to education and training. The percentage of IT budget allocated to education and training is on the increase across all of the geographies, which can be an indication that organizations are willing to spend more money to prepare and retain their talent. It is also an indication of the current state of affairs,

where short stay of talents are preferable to long stay of stales; as an IT manager of one of the fastest growing businesses in Australia put it during a recent conversation with one of the authors: 'I prefer to spend money on training my staff even if it means they leave, rather than not training them and they hang around'.

Training budget projections for 2011 is around 4% in the US and SE Asia, 3.2% in Europe, and a very strong 5.8% in Latin America. Owing to the high priority placed on a balanced set of skills required (Luftman and Ben-Zvi, 2010a), there is an anticipated growth in education and training focused on interpersonal (e.g., writing, presentation, and negotiation skills, and ethics/morals), and business/management skills, followed by training in high priority technologies such as business intelligence, virtualization, and cloud computing.

Lastly, with regard to the rate of IT staff turnover there are surprisingly good staff retention rates across the globe during the economic downturn. Staff turnover was 5.82% in Europe, 5.88% in SE Asia, and 5.49% in Latin America. In the US staff turnover was 5.5% (consistently decreasing from 7.2% in 2006).

D. IT organization considerations

This section discusses the survey findings related to IT organizational structure, CIOs, and IT outsourcing.

1. IT organization structure

The organization of IT can have a major impact on the performance of the company. IT organization structure is the degree to which it is centralized, decentralized, or federalized. In all, 65.6% of European, 45% of SE Asians, 69% of Latin American, and 68% of the US respondents indicated centralized IT organization structure, where all IT reports to a single IT executive (the CIO). Centralized IT organization structure can better attain more IT standardization across the organization; for example, a centralized email system ensures the same email features across the enterprise (same look and feel, same capabilities, centralized archive, and back up policies, etc.).

On the other end of the spectrum, there are decentralized organizational structures where each business unit has its own IT organization without much coordination across business units (e.g., each unit having their own email system, or their own standards for database administration). There is not much economy of scale in the decentralized structure, but each business unit has the full flexibility to focus on the unit's particular IT needs (applications and infrastructure). This can be of most value in large organizations where IT needs across some business units vary greatly. Some universities, for example, can benefit from decentralized IT structure, as each school or department might have vastly different IT requirements. In all, 9.8% of the European, 15% of the SE Asian, and 9% of Latin American companies indicated a decentralized IT structure. Interestingly, only 2% of US respondents indicated a decentralized IT structure, a big drop from 9.5% in 2009 (Luftman and Ben-Zvi, 2010a). This drop could be the result of a radical response to the recession, in which many of the US CIOs have opted for economies of scale over

flexibility of business units in order to rein in their IT expenditures.

A federated (or hybrid) structure can realize the benefits from both centralized and decentralized structures. Corporate-wide standards are enforced in an effort to maximize the benefits of economies of scale, while providing flexibility to business units to maximize unique application opportunities at the business unit level. In all, 23% of the European companies, 25% of the SE Asian, and only 14% of Latin American companies were reported as federated in 2010. Organizations with a federated structure tend to have a higher alignment maturity assessment than those that are centralized or decentralized (Luftman *et al.*, 2010). Therefore, it should not come as a surprise to see these numbers increase over the next few years (in the US, e.g., percentage of companies with hybrid IT organization structure rose from 18.4% in 2009 to 28% in 2010).

E. CIO trends

1. CIO reporting structure and role of CIO

As the majority of time of CIOs is spent in dealing with non-technical issues (see Table 6; discussed below), the roles of CIOs vary between the geographies surveyed.

Table 7 shows to whom do CIOs or senior IT executives report to. Previous research has shown that, on average, organizations in which CIOs report to CEOs have higher alignment maturity than those reporting to non-CEO executives (Luftman and Ben-Zvi, 2010b; Luftman *et al.*, 2010). CIO reporting to the CEO is the highest in SE Asia (68%). US and Europe come in as distant third and fourth (44% and 43%, respectively). But Latin America comes in at a distant last with only 13% of CIOs reporting directly to their CEO.

2. CIO tenure

The average CIO tenure is on the rise in all geographies. This trend has been clearly illustrated in previous SIM (US-based) surveys with the average CIO tenure of 4.6 years in 2009, 4.3 years in 2008, 4.1 years in 2007, and 3.6 years in 2006. In 2010, the respondents reported 52.3% in SE Asian CIOs held their positions for more than 3 years, followed by 52% in the US, 51% in Europe, and 48% in Latin America.

Even though it is believed that high CIO turnover (short tenure) makes it difficult for CIOs to address any long-term changes to the business or IT organization (Luftman and Ben-Zvi, 2010b), other research shows executive performance peaks between 4 and 5 years of tenure, after which the performance is likely to dip before another performance increase for executives with tenure of more than 8 years (O'Shannassy, 2011). The same research shows the optimum executive tenure to be either between 4 and 6 years, or tenures longer than 8 years (O'Shannassy, 2011). It is noted that this recent study was for CEOs and board of directors, with an average tenure of 8 years, and not specifically for CIOs. A similar study focusing on IT executives would be an interesting and welcome addition to the body of literature in this area.

The survey also asked respondents to indicate where CIOs were hired from. In all, 58% of the US and European,

Table 6 Activities that CIOs Spend Time On

	<i>Global (average) (%)</i>	<i>US (%)</i>	<i>Europe (%)</i>	<i>SE Asia (inc Australia) (%)</i>	<i>Latin America (%)</i>
Architecture	9	7	7	12	10
Human resources	8	7	9	5	9
IT governance	10	9	11	10	11
Non-IT	10	11	10	11	8
Operations	12	12	11	13	13
Relationship management with business	14	18	17	11	9
Relationship management with IT staff	10	11	12	12	6
Relationship management with vendors	7	6	10	4	6
Software development	8	6	4	7	14
Strategy	12	13	9	15	10

Table 7 CIO or Senior IT Executive Reporting

<i>IT Executives Report to:</i>	<i>Global (average) (%)</i>	<i>US (%)</i>	<i>Europe (%)</i>	<i>SE Asia (inc Australia) (%)</i>	<i>Latin America (%)</i>
CEO	42	44	43	68	13
CFO	19	31	29	10	6
Board of directors	9		9	12	6
Business unit executive	21	4	7	5	66
Other corporate executive	9	10	12	5	9

42% of SE Asian, and 36% of Latin American respondents said their CIO was hired from outside the company. In other words, 42% of the US and European, 58% SE Asian, and 64% of Latin American CIOs were hired from within the company; hence unless you are in SE Asia or Latin America the odds of getting a CIO position in your current company is not in your favor.

When asked what background CIOs had before taking up their current position, the respondent indicated that in the US, 92% of CIOs were recruited from IT organizations (either internal or external). The same figure was 82% for Europe and 70% for Latin America, and also very high for SE Asia, with 91% of CIOs being recruited from IT organizations. In other words, if you do not have an IT background, and you are eyeing a CIO position, the best places to be are Latin America and Europe, but even then the chances of landing that CIO job, without an IT background, are not very high.

3. CIO time on activities

Predictably, CIOs spend most of their time dealing with non-technical issues; 75% in Europe and the US, 68% in SE Asia, and 57% in Latin America. While IT leaders in Australia spend almost half of their time on governance, operations, and strategy, their Latin American and SE Asian counterparts spend only about one-third of their time on similar issues. Interestingly, CIO-time spent on software development issues is around 4%–6% across all the geographies barring Latin America where it accounts for 14% of CIOs' time.

Because of the relatively less mature infrastructure in SE Asia and Latin America, it is expected that IT leaders in

those areas spend more time on technical issues than their counterparts in the US and Europe. Indeed, while US CIOs spend 19% of their time on operations and architecture (and European CIOs even less, at only 18%), the same activities take up 25% of the time of IT leaders in SE Asia, 23% in Latin America. We expect these numbers to move closer to that of the US and Europe as the infrastructure in those areas matures.

In an interesting departure from the other areas, some SE Asian CIOs (less than 4%) indicated they spend some of their time on philanthropy and on research and development.

F. IT outsourcing

IT outsourcing has been defined as 'handing over the management of some or all of an organization's IT assets, resources and related services to a third party' (Willcocks *et al.*, 1995). Over the last two decades, IT outsourcing has not only been an attractive option for many corporations, but also subject of much academic research (Dibbern *et al.*, 2004; Gonzalez *et al.*, 2006; Lacity *et al.*, 2009).

IT leaders globally have long been looking to outsourcing as a mean to reduce costs as well as to fill skills gaps. The recent recession has fueled this even further; the overall increase in outsourcing in all of the areas being a testimony to that. Considering total outsourcing, which includes offshoring, near-shoring, and consulting of non-internal staff, the allocation of the respective IT budgets are 37% in Europe, 23% in SE Asia, 12% in Latin America, and 22% in the US.

In the US, outsourced offshore staff are set to increase from 5% of total IT budget in 2010 to 7% in 2011. The same

is true in Europe (5% of IT budget in 2010 to 6% in 2011), SE Asia (1.5% to 2%). The reason the increase in outsourced offshore staffing in SE Asia is more modest than all of the other areas is due to the number of high profile offshoring failures in the early 2000s (e.g., see Gray, 2003); in some cases, public backlash against organizations with extensive outsourcing offshore staff has been hard to ignore (NineNews, 2006; Dubie, 2009; Gluyas, 2009). The only exception to this pattern is Latin America, where outsourced offshore staff is set to remain flat (at 1%). This is most probably due to a massive increase in internal offshore staff in that geography, which is set to increase from 16% in 2010 to 34% in 2011.

Outsourced domestic staff also mostly follow the same general pattern, increasing from 20% of IT budget in 2010 to 22% in 2011 in Europe, from 4.2% to 4.7% in SE Asia, from 7% to 9% in Latin America, and from 7% to 8% in the US.

Like other outsourcing, US IT leaders predict an increase in consulting services expenses in 2011 (from 10% to 11% of total IT budget), while it will be reduced in SE Asia and Latin America to 10% and a mere 2%, respectively. In Europe it is anticipated that consulting services remain flat at 12%.

It would be remiss not to highlight the departure from the norm in Europe for outsourced domestic staff. Outsourced domestic staffing in Europe stands at a staggering 20%, compared to 7% in all the other demographics. With the free movement of the labor force in the European Union (E.U., 2005), many European enterprises opt for employing outsourced staff from the neighboring countries (E.U., 2009; Gasteen, 2010). For example, the recent ruling of the European Court of Justice lifting the limit on the use of foreign workers in Germany (Slegers, 2010), labor employment across borders has become more streamlined. With this ruling, outsourced domestic employment in Europe is set to increase even higher than the current level. Our survey predicts a 10% increase in 2011, jumping to 22% (compared to 20% in 2010). Hence, due to the current employment legislations in Europe, the anticipated increase in outsourcing is mainly driven through increased outsourced domestic staffing.

The projected rise in overall outsourcing in the US, Europe, SE Asia, and Latin America is related to organizations looking to reduce costs as well as to fill skills gaps. Overall, the prognosis for outsourcing is that it will continue to rise, at least in the near term in all the geographies.

Concluding remarks

The rather slow recovery from the recent recession poses new challenges to IT executives around the globe. The relatively consistent top managerial concerns cannot simply be addressed through identical responses in different geographies; each area has its own set of characteristics and appropriate response to management concerns. In other words, unique characteristics of the *local markets* influence management responses of enterprises operating in a *globally linked* environment. While the recovery from the recession presents new challenges and opportunities for IT executives in 2011, IT executives are set to leverage both *global* and *local* IT opportunities (such as increased

spending and hiring, business intelligence, virtualization, and outsourcing) to overcome not only global challenges but also local IT and business challenges.

By comparing and contrasting different geographies, this research has identified the many similarities and dissimilarities that confront managers. Clearly, there are regional influences that are powerful enough to reduce the influence of global trends.

In closing, it is important to point out that IT managers are working in a highly inter-connected world, and therefore certain patterns in different geographic locations are evident. However, this research found that while there are many similarities there are also important local trends that managers must be sensitive to.

The authors envision extending the survey's participation to all geographies and welcome proposals to assist in obtaining additional data.

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About the authors

Jerry Luftman's career includes strategic positions in management (Information Technology and consulting), management consulting, Information Systems, and education. Dr. Luftman's experience combines the strengths of practitioner, consultant, and academic. His proficiency in business-IT alignment, 18 books, published research, consulting, mentoring, and teaching/speaking engagements further exemplify Dr. Luftman's expertise and leadership in his field. After a notable 22-year career with IBM, he is currently a Distinguished Professor at Stevens Institute of Technology School of Management. In addition to the traditional responsibilities of a Professor, teaching Master's and Ph.D. students, Dr. Luftman is the founder and Associate Dean of the Stevens' Information Systems programs, one of the largest in the world. In this capacity, Dr. Luftman manages all aspects of the program including course building, research, marketing, administrative processes, and executive programs. Dr. Luftman's project experience ranges from senior management issues through tactical implementation. His current initiatives include creating business and information technology strategies and plans, facilitating IT governance processes, and developing and deploying applications leveraging emerging technologies. Dr. Luftman most recently pioneered the vehicle for assessing the maturity of IT-business alignment, where he has a repository of over one-third of the Global 1000 companies. He also serves on the executive board

of several companies, organizations, and journals. Dr. Luftman's last responsibility at IBM was as a key speaker at their Customer Executive Conference Center in Palisades, NY. While responsible for management research, he played a significant role in defining and introducing the IT Strategy and Planning and Business Process Reengineering practice methods for the IBM Management Consulting Group. His framework for applying the strategic alignment model is fundamental in helping clients understand, define, and scope the appropriate strategic planning approach to pursue. Dr. Luftman's annual international IT trends survey sponsored by SIM is recognized internationally as an industry barometer.

Hossein S. Zadeh's career includes positions as research scientist, senior IT manager, senior project manager, management consulting, system analyst, and educator. Dr. Zadeh's experience combines disciplines of mechanical and aerospace engineering on the one hand and management and information systems on the other. He has researched and published in fields as diverse as maneuver control of satellites to the innovative applications of information systems in healthcare. After a management experience looking after a large-scale IT department with 15,000+ clients, Hossein now holds the position of tenured Senior Lecturer and Program Director in the School of Business Information Technology and Logistics, RMIT University (Melbourne, Australia). His research, linking the diverse fields of engineering, management, and IT has attracted over \$1,000,000 in grants, awards, scholarships, and contracts, from organizations such as the Australian Research Council and the Australian Department of Defence. In 2004, Hossein was a Visiting Scholar at Linköping University, Sweden, and in 2009/2010, was a Distinguished Visiting Scholar at IBM Almaden Research Labs, San Jose, USA. Hossein has designed undergraduate and graduate programs and courses and has taught undergraduate and graduate courses in Australia, Hong Kong, Vietnam, Singapore, and Sweden. Hossein is the recipient of the 2008 University Team Teaching Award and 2009 University Certificate of Achievement in innovative teaching. In 2005, Hossein was director of undergraduate studies in the School of Business Information Technology, RMIT University, one of the largest of such schools in Australia with more than 3000 students and more than 70 tenured academic posts. Hossein's current research focuses on IT services and healthcare services as pillars of most modern (service) economies. He is a continuing reviewer of

multiple international journals and conferences, was associate editor of IT Services track in ICIS 2010 and is mini-track chair of AmCIS 2011, and has been a session organizer and reviewer of IEEE Aerospace conference since 2002. In 2010, Hossein was a recipient of the prestigious and internationally recognized IBM Faculty Award.

Appendix

Survey methods

The SIM survey has been conducted since 1980. Surveys prior to 2000 focused just on the top management concerns. Since 2003, the survey has been extended to pursue more specific insights regarding the key IT issues of the day. A significant strength of this research is in its ability to identify important trends by comparing survey data from previous years. Beginning in 2008 the survey has been extended to IT executives from around the globe.

The 2010 survey was similar to previous ones in methodology and process. The questions were based on previous surveys, with questions modified based on previous results and suggestions from respondents and researchers (academic and industry). In addition, some questions were updated and new questions were added based on (1) lists from other similar research, (2) input from Board members from sponsoring organizations, and (3) the lead author's experience.

SIM members represent the US respondents, while CIO net members represent the European respondents. The Asian results were facilitated by Hossein S. Zadeh (co-author), and those of Latin America were facilitated by Martin Santana (ESAN) and Guillermo Rodriguez (UNAM). Senior IT executives were invited to take the online survey. The key US findings were initially presented during the 2010 SIM annual conference (SIMposium) in Atlanta, GA in late September 2010, and global findings have been presented throughout the fourth quarter and first quarter of 2011.

The US findings and initial international findings have been published in the fourth quarter issue of *MISQE*. The purpose of the *JIT* paper is to provide important international insights.

The authors anticipate extending the reach of this important research to a more complete set of countries and geographies, and invite leading researchers with a strong network of IT executives to contact us.